

India is my country and all Indians are my brothers and sisters. I love my country and I am proud of its rich and varied heritage. I shall always strive to be worthy of it. I shall give my parents, teachers, and all elders respect and treat everyone with courtesy. To my country and my people, I pledge my devotion. In their well-being and prosperity alone, lies my happiness.



भारत मेरा देश है। सब भारतवासी मेरे भाई-बहन हैं। मैं अपने देश से प्रेम करता/करती हूँ। इसकी समृद्ध एवं विविध संस्कृति पर मुझे गर्व है। मैं सदैव इसका सुयोग्य अधिकारी बनने का प्रयत्न करता/करती रहूँगा/रहूँगी। मैं अपने माता-पिता, शिक्षकों एवं गुरुजनों का सम्मान करूँगा/करूँगी और प्रत्येक के साथ विनय रखूँगा/रहूँगी। मैं अपने देश और देशवासियों के प्रति सत्यनिष्ठा की प्रतिज्ञा करता/करती हूँ। इनके कल्याण एवं समृद्धि में ही मेरा सुख निहित है।

Home Work

Keyvault ka code bana ke lana hai

Post creation of VM, Terraform and nginx should be installed on all machines using provisioners / custom data script

Recap

Phase1: Windows and Linux VM pr one tier app ka deployment kia

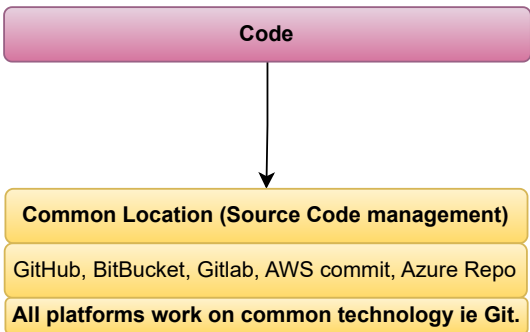
Phase2: Azure Landing Zone-> Manual

Hierarchy , Networking, Compute, Loadbalancing, Bastion, Database, NSG

Phase2: Azure Landing Zone-> Terraform

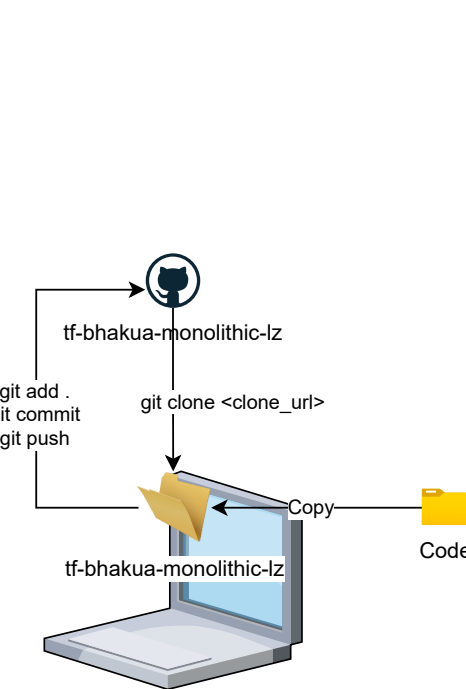
Child and Parent Module ke through kia for_each, map of objects, dynamic block, conditional iteration, output block, data block, lifecycle, remote backend, custom data script, Terraform Functions, provider, locking, import block, null resource, terraform taint

Terraform ke through azure landing zone child parent module best practices ke saath banana sikh gae jisme state file ko remote backend storage account ke andar humne isko rakha

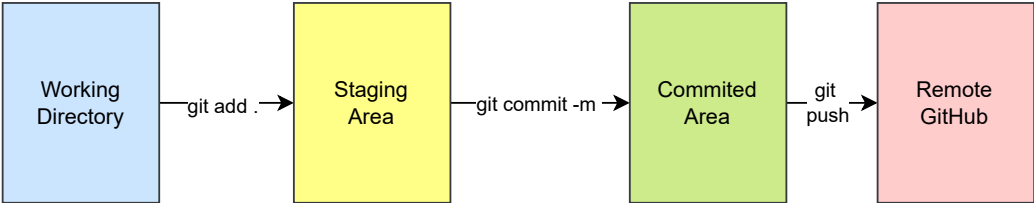
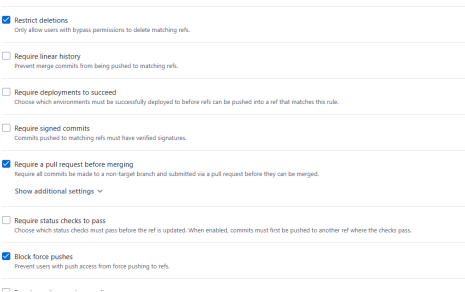


Source Code management Benefits
1. Stores Version History of Code
2. Code ka Backup
3. Multiple Teammates kaam kr skte hai saath me
4. Conflict Management
5. Rollback
6. Branch Isolation
7. Code Review

- Jo Code local pr rakha hai, yaha se utha ke remote github pr leke jana hai and branching strategy ke saath rakhna hai
1. Create a new **GitHub Account**
2. Login to Github Account and Create Repository
3. Clone Repository
4. Create feature branch banaenge and Code will be pushed
5. Raise PR from feature to main branch
6. PR will be reviewed by the reviewer and approved



1. Create GitHub Account
2. Create Public Repository
- Enable README, Gitignore
3. Install Git in Computer
4. Clone Repository to Local
- git clone <Clone URL>
5. Verify Repository
- git branch, git status
6. Matched - Nothing to Commit
7. Module and Environment Folder ko cut karke teju-bhaiya folder me gae aur paste kar dia same level pr jaha readme and gitignore tha
8. Again git status
New folders added new, inko add krne ke lie git add chalo
- git ke track me kisi file ko lane ke lie git add . use krte hai
9. git commit -m "new commit"
10. git push
11. Protect Main branch with Protection Rule
- github repo ki setting me jake main branch pr policy lgaenge
12. Changes push krne ke lie feature branch banana pdega main branch se
- git branch = show current branch
- git checkout -b feature/new-rg = to create new branch



Kabhi bhi main branch me directly push nahi krna hai

Requirement - New RG add karna hai
1. Create feature branch banaya and changes pushed
2. Automatic Pipeline run in github action
Scan Stage (tfsec, gitleaks, tfint, infracost)
Plan Stage (terraform init, fmt, validate, plan)
3. Raise PR to merge feature to main branch
4. PR will be reviewed by Human Reviewer
5. Human will check if everything is fine from feature code, scan report and plan output
6. Human will approve the code changes
7. The Code will be merged to main branch and automatic pipeline will run from main branch
Scan -> Plan -> Approval(Only on main) -> Apply(Only on main)
Best Practices
Never Commit directly to the main branch, Always raise PR to merge to main brach
Write always meaningful commit messages
Keep PR short and focused
Always enable branch policy
Keep Atleast 2 reviewers
Store Secrets in GitHub Secrets
Always use Tags for production
Delete merged feature branch to keep repository clean